# Integration Manual

for S32K3 GPT Driver

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# Chapter 1

# Revision History

| Revision | Date | Author | Description |
|---|---|---|---|
| 1.0 | 31.03.2022 | NXP RTD Team | Prepared for release RTD S32K3 2.0.0 |

# Chapter 2

# Introduction


- Supported Derivatives
- Overview
- About This Manual
- Acronyms and Definitions
- Reference List


This integration manual describes the integration requirements for GPT Driver for S32K3 microcontrollers.

## 2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k344_mapbga257
- s32k344_mqfp172
- s32k324_mapbga257
- s32k324_mqfp172
- s32k314_mapbga257
- s32k314_mqfp172
- s32k312_mqfp172
- s32k312_mqfp100
- s32k322_mqfp172
- s32k322_mqfp100
- s32k342_mqfp172
- s32k342_mqfp100
- s32k341_mqfp172
- s32k341_mqfp100

All of the above microcontroller devices are collectively named as S32K3.

## 2.2 Overview

**AUTOSAR (AUTomotive Open System ARchitecture)** is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

## 2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

## 2.4 Acronyms and Definitions

| Term | Definition |
|---|---|
| API | Application Programming Interface |
| ASM | Assembler |
| BSMI | Basic Software Make file Interface |
| GPT | General Purpose Timer |
| C/CPP | C and C++ Source Code |
| CS | Chip Select |
| CTU | Cross Trigger Unit |
| DEM | Diagnostic Event Manager |
| DET | Development Error Tracer |
| DMA | Direct Memory Access |
| ECU | Electronic Control Unit |
| FIFO | First In First Out |
| LSB | Least Significant Bit |
| MCU | Micro Controller Unit |
| OS | Operating System |
| MIDE | Multi Integrated Development Environment |
| MSB | Most Significant Bit |
| N/A | Not Applicable |
| RAM | Random Access Memory |
| SIU | Systems Integration Unit |
| SWS | Software Specification |
| VLE | Variable Length Encoding |
| XML | Extensible Markup Language |

## 2.5 Reference List

| # | Title | Version |
|---|---|---|
| 1 | Specification of GPT Driver | AUTOSAR Release 4.4.0 |
| 2 | S32K3XX Reference Manual | S32K3xx Reference Manual, Rev. 3, 10/2021 |
| 3 | Datasheet | S32K3xx Data Sheet, Rev. 3 — 10/2021 |
| 4 | S32K342 Errata | S32K342, Mask Set Errata for Mask 0P97C, Rev. 13 Dec 2021 |
| 5 | S32K312 Errata | S32K312, Mask Set Errata for Mask 0P09C, Rev. 10/Jan/2022 |
| 6 | S32K344 Errata | S32K344, Mask Set Errata for Mask 0P55A/1P55A, Rev. 30/Sep/2021 |

# Chapter 3

# Building the driver

- Build Options
- Files required for compilation
- Setting up the plugins

This section describes the source files and various compilers, linker options used for building the driver.
It also explains the EB Tresos Studio plugin setup procedure.

## 3.1 Build Options

- GCC Compiler/Assembler/Linker Options
- GHS Compiler/Assembler/Linker Options
- IAR Compiler/Assembler/Linker Options

The RTD driver files are compiled using:

- NXP GCC 10.2.0 20200723 (Build 1728 Revision g5963bc8)
- IAR ANSI C/C++ Compiler V8.50.10.290/W32 for ARM Functional Safety
- Green Hills Multi 7.1.6d / Compiler 2021.1.4

The compiler, assembler, and linker flags used for building the driver are explained below.
The TS_T40D34M20I0R0 part of the plugin name is composed as follows:

- T = Target_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative_Id (e.g. D34 identifies S32K3 platform)
- M = SW_Version_Major and SW_Version_Minor
- I = SW_Version_Patch
- R = Reserved

### 3.1.1 GCC Compiler/Assembler/Linker Options

#### 3.1.1.1 GCC Compiler Options

| Compiler Option | Description |
|---|---|
| -mcpu=cortex-m7 | Targeted ARM processor for which GCC should tune the performance of the code |
| -mthumb | Generates code that executes in Thumb state |
| -mlittle-endian | Generate code for a processor running in little-endian mode |
| -mfpu=fpv5-sp-d16 | Specifies the floating-point hardware available on the target |
| -mfloat-abi=hard | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions |
| -std=c99 | Specifies the ISO C99 base standard |
| -Os | Optimize for size. Enables all -O2 optimizations except those that often increase code size |
| -ggdb3 | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -Wall | Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros |
| -Wextra | This enables some extra warning flags that are not enabled by -Wall |
| -pedantic | Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioend -std option |
| -Wstrict-prototypes | Warn if a function is declared or defined without specifying the argument types |
| -Wundef | Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero |
| -Wunused | Warn whenever a function, variable, label, value, macro is unused |
| -Werror=implicit-function-declaration | Make the specified warning into an error. This option throws an error when a function is used before being declared |
| -Wsign-compare | Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned. |
| -Wdouble-promotion | Give a warning when a value of type float is implicitly promoted to double |
| -fno-short-enums | Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values. |
| -funsigned-char | Let the type char be unsigned by default, when the declaration does not use either signed or unsigned |
| -funsigned-bitfields | Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned |
| -fomit-frame-pointer | Omit the frame pointer in functions that dont need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available. |

| Compiler Option | Description |
|---|---|
| -fno-common | Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit |
| -fstack-usage | Makes the compiler output stack usage information for the program, on a per-function basis |
| -fdump-ipa-all | Enables all inter-procedural analysis dumps |
| -c | Stop after assembly and produce an object file for each source file |
| -DS32K3XX | Predefine S32K3XX as a macro, with definition 1 |
| -DS32K344 | Predefine S32K344 as a macro, with definition 1 |
| -DGCC | Predefine GCC as a macro, with definition 1 |
| -DUSE_SW_VECTOR_MODE | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode |
| -DD_CACHE_ENABLE | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.↩ c under the Platform driver |
| -DI_CACHE_ENABLE | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver |
| -DENABLE_FPU | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPO↩ RT as a macro, with definition 1. Allows drivers to be configured in user mode. |

#### 3.1.1.2 GCC Assembler Options

| Assembler Option | Description |
|---|---|
| -Xassembler-with-cpp | Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix) |
| -mcpu=cortexm7 | Targeted ARM processor for which GCC should tune the performance of the code |
| -mthumb | Generates code that executes in Thumb state |
| -c | Stop after assembly and produce an object file for each source file |

#### 3.1.1.3 GCC Linker Options

| Linker Option | Description |
|---|---|
| -Wl,-Map,filename | Produces a map file |
| -T linkerfile | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it) |
| –entry=Reset_Handler | Specifies that the program entry point is Reset_Handler |

| Linker Option | Description |
|---|---|
| -nostartfiles | Do not use the standard system startup files when linking |
| -mcpu=cortexm7 | Targeted ARM processor for which GCC should tune the performance of the code |
| -mthumb | Generates code that executes in Thumb state |
| -mfpu=fpv5-sp-d16 | Specifies the floating-point hardware available on the target |
| -mfloat-abi=hard | Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions |
| -mlittle-endian | Generate code for a processor running in little-endian mode |
| -ggdb3 | Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program |
| -lc | Link with the C library |
| -lm | Link with the Math library |
| -lgcc | Link with the GCC library |

### 3.1.2 GHS Compiler/Assembler/Linker Options

#### 3.1.2.1 GHS Compiler Options

| Compiler Option | Description |
|---|---|
| -cpu=cortexm7 | Selects target processor: Arm Cortex M7 |
| -thumb | Selects generating code that executes in Thumb state |
| -fpu=vfpv5_d16 | Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers |
| -fsingle | Use hardware single-precision, software double-precision FP instructions |
| -C99 | Use (strict ISO) C99 standard (without extensions) |
| –ghstd=last | Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++) |
| -Osize | Optimize for size |
| –gnu_asm | Enables GNU extended asm syntax support |
| -dual_debug | Generate DWARF 2.0 debug information |
| -G | Generate debug information |
| -keeptempfiles | Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory |
| -Wimplicit-int | Produce warnings if functions are assumed to return int |
| -Wshadow | Produce warnings if variables are shadowed |
| -Wtrigraphs | Produce warnings if trigraphs are detected |
| -Wundef | Produce a warning if undefined identifiers are used in #if preprocessor statements |
| –unsigned_chars | Let the type char be unsigned, like unsigned char |

| Compiler Option | Description |
|---|---|
| –unsigned_fields | Bitfelds declared with an integer type are unsigned |
| –no_commons | Allocates uninitialized global variables to a section and initializes them to zero at program startup |
| –no_exceptions | Disables C++ support for exception handling |
| –no_slash_comment | C++ style // comments are not accepted andgenerate errors |
| –prototype_errors | Controls the treatment of functions referenced or called when no prototype has been provided |
| –incorrect_pragma_warnings | Controls the treatment of valid #pragma directives that use the wrong syntax |
| -c | Stop after assembly and produce an object file for each source file |
| -DS32K3XX | Predefine S32K3XX as a macro, with definition 1 |
| -DS32K344 | Predefine S32K344 as a macro, with definition 1 |
| -DGHS | Predefine GHS as a macro, with definition 1 |
| -DUSE_SW_VECTOR_MODE | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode |
| -DD_CACHE_ENABLE | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.↩ c under the Platform driver |
| -DI_CACHE_ENABLE | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver |
| -DENABLE_FPU | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPO↩ RT as a macro, with definition 1. Allows drivers to be configured in user mode |

#### 3.1.2.2 GHS Assembler Options

| Assembler Option | Description |
|---|---|
| -cpu=cortexm7 | Selects target processor: Arm Cortex M7 |
| -preprocess_assembly_files | Controls whether assembly files with standard extensions such as .s and .asm are preprocessed |
| -list | Creates a listing by using the name and directory of the object file with the .lst extension |
| -c | Stop after assembly and produce an object file for each source file |

#### 3.1.2.3 GHS Linker Options

| Linker Option | Description |
|---|---|
| -e Reset_Handler | Make the symbol Reset_Handler be treated as a root symbol and the start label of the application |

| Linker Option | Description |
|---|---|
| -T linker_script_file.ld | Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it) |
| -map | Produce a map file |
| -keepmap | Controls the retention of the map file in the event of a link error |
| -Mn | Generates a listing of symbols sorted alphabetically/numerically by address |
| -delete | Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them |
| -ignore_debug_references | Ignores relocations from DWARF debug sections when using -delete. DWA↩ RF debug information will contain references to deleted functions that may break some third-party debuggers |
| -Llibrary_path | Points to library_path (the libraries location) for thumb2 to be used for linking |
| -larch | Link architecture specific library |
| -lstartup | Link run-time environment startup routines. The source code for themodules in this library is provided in the src/libstartup directory |
| -lind_sd | Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library |
| -v | Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols |
| -keep=C40_Ip_AccessCode | Avoid linker remove function C40_Ip_AccessCode from Fls module because it is not referenced explicitly |
| -nostartfiles | Controls the start files to be linked into the executable |

### 3.1.3 IAR Compiler/Assembler/Linker Options

#### 3.1.3.1 IAR Compiler Options

| Compiler Option | Description |
|---|---|
| –cpu=Cortex-M7 | Targeted ARM processor for which IAR should tune the performance of the code |
| –cpu_mode=thumb | Generates code that executes in Thumb state |
| –endian=little | Generate code for a processor running in little-endian mode |
| –fpu=FPv5-SP | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. |
| -e | Enables all IAR C language extensions |
| -Ohz | Optimize for size. the compiler will emit AEABI attributes indicating the requested optimization goal. This information can be used by the linker to select smaller or faster variants of DLIB library functions |
| –debug | Makes the compiler include debugging information in the object modules. Including debug information will make the object files larger |

| Compiler Option | Description |
|---|---|
| –no_clustering | Disables static clustering optimizations. Static and global variables defined within the same module will not be arranged so that variables that are accessed in the same function are close to each other |
| –no_mem_idioms | Makes the compiler not optimize certain memory access patterns |
| –do_explicit_zero_opt_in_named_sections | Disable the exception for variables in user-named sections, and thus treat explicit initializations to zero as zero initializations, not copy initializations |
| –require_prototypes | Force the compiler to verify that all functions have proper prototypes. Generates an error otherwise |
| –no_wrap_diagnostics | Does not wrap long lines in diagnostic messages |
| –diag_suppress=Pa050 | Suppresses diagnostic message Pa050 |
| -DS32K3XX | Predefine S32K3XX as a macro, with definition 1 |
| -DS32K344 | Predefine S32K344 as a macro, with definition 1 |
| -DIAR | Predefine IAR as a macro, with definition 1 |
| -DUSE_SW_VECTOR_MODE | Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode. |
| -DD_CACHE_ENABLE | Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initalization in source file system.↩ c under the Platform driver |
| -DI_CACHE_ENABLE | Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initalization in source file system.c under the Platform driver |
| -DENABLE_FPU | Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initalization in source file system.c under the Platform driver |
| -DMCAL_ENABLE_USER_MODE_SUPPORT | Predefine MCAL_ENABLE_USER_MODE_SUPPO↩ RT as a macro, with definition 1. Allows drivers to be configured in user mode. |

#### 3.1.3.2 IAR Assembler Options

| Assembler Option | Description |
|---|---|
| –cpu Cortex-M7 | Targeted ARM processor for which IAR should generate the instruction set |
| –cpu_mode thumb | Selects the thumb mode for the assembler directive CODE |
| -g | Disables the automatic search for system include files |
| -r | Generates debug information |

#### 3.1.3.3 IAR Linker Options

| Linker Option | Description |
|---|---|
| –map filename | Produces a map file |
| –config linkerfile | Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it) |

| Linker Option | Description |
|---|---|
| –cpu=Cortex-M7 | Selects the ARM processor variant to link the application for |
| –fpu=FPv5-SP | Use this option to generate code that performs floating-point operations using a Floating Point Unit (FPU). Single-precision variant. |
| –entry __start | Treats __start as a root symbol and start label |
| –enable_stack_usage | Enables stack usage analysis. If a linker map file is produced, a stack usage chapter is included in the map file |
| –skip_dynamic_initialization | Dynamic initialization (typically initialization of C++ objects with static storage duration) will not be performed automatically during application startup |
| –no_wrap_diagnostics | Does not wrap long lines in diagnostic messages |

## 3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the GPT driver for S32K3 microcontrollers. To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

**GPT** files:

- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0\src\Gpt.c
- ..\Gpt_TS_T40D34M20I0R0\src\Gpt_Ipw.c
- ..\Gpt_TS_T40D34M20I0R0\src\Pit_Ip.c
- ..\Gpt_TS_T40D34M20I0R0\src\Stm_Ip.c

- ..\Gpt_TS_T40D34M20I0R0\src\Rtc_Ip.c
- ..\Gpt_TS_T40D34M20I0R0\src\Emios_Gpt_Ip.c

**GPT** generated files (these files should be generated by the user using a configuration tool):

- Gpt_Cfg.h
- Pit_Ip_Cfg.h
- Pit_Ip_Cfg_Defines.h
- Rtc_Ip_Cfg.h
- Rtc_Ip_Cfg_Defines.h
- Stm_Ip_Cfg.h
- Stm_Ip_Cfg_Defines.h
- Emios_Gpt_Ip_Cfg.h
- Emios_Gpt_Ip_Cfg_Defines.h
- Gpt_Ipw_PBcfg.h
- Gpt_PBcfg.h
- Pit_Ip_PBcfg.h
- Rtc_Ip_PBcfg.h
- Stm_Ip_PBcfg.h
- Emios_Gpt_Ip_PBcfg.h
- Gpt_Cfg.c
- Gpt_PBcfg.c
- Pit_Ip_PBcfg.c
- Rtc_Ip_PBcfg.c
- Stm_Ip_PBcfg.c
- Emios_Gpt_Ip_PBcfg.c

**Files** from Base common folder:

- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0.h

- ..\Base_TS_T40D34M20I0R0.h
- ..\Base_TS_T40D34M20I0R0\generate_PC.h
- ..\Base_TS_T40D34M20I0R0\header\S32K344.h

**Files** from Det folder:

- ..\Det_TS_T40D34M20I0R0.h

## 3.3 Setting up the plugins

The GPT driver was designed to be configured by using the EB Tresos Studio (version EB tresos Studio 27.1.0 or later).

### 3.3.1 Location of various files inside the Gpt module folder

VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:

- ..\Gpt_TS_T40D34M20I0R0\config\Gpt.xdm
- ..\Base_TS_T40D34M20I0R0\config\Base.xdm
- ..\Resource_TS_T40D34M20I0R0\config\Resource.xdm VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
- ..\Gpt_TS_T40D34M20I0R0\autosar\Gpt.epd
- ..\Base_TS_T40D34M20I0R0\autosar\Base.epd
- ..\Resource_TS_T40D34M20I0R0\autosar\Resource.epd Code Generation Templates for Pre-Compile time configuration parameters:
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0.h
- ..\Gpt_TS_T40D34M20I0R0\src\Gpt_Cfg.c

### 3.3.2 Dependencies

- RESOURCE is required to select processor derivative. Current Gpt driver has support for the following derivatives, each one having attached a Resource file: s32k312_mqfp100, s32k312_mqfp172, s32k314_mapbga257, s32k314_mqfp172, s32k322_mqfp100, s32k322_mqfp172, s32k324_mapbga257, s32k324_mqfp172, s32k341↩_mqfp100, s32k341_mqfp172, s32k342_mqfp100, s32k342_mqfp172, s32k344_mapbga257, s32k344_mqfp172
- DET is required for signaling the development error detection (parameters out of range, null pointers, etc).
- BASE is required for Gpt specific header files and other header definitions.

# Chapter 4

# Function calls to module


- Function Calls during Start-up
- Function Calls during Shutdown
- Function Calls during Wake-up


## 4.1 Function Calls during Start-up

GPT shall be initialized during STARTUP1 phase of EcuM initialization. The API to be called for this is Gpt_Init() MCU module shall be initialized before GPT is initialized.

## 4.2 Function Calls during Shutdown

If GptWakeupFunctionalityApi and GptWakeupSourceRef are enabled, Gpt_SetMode(GPT_MODE_SLEEP) API shall be called during GO SLEEP phase of EcuM to configure the hardware for Sleep mode.

## 4.3 Function Calls during Wake-up

For the platforms where the GPT driver controls wakeup hw sources, if the GptWakeupFunctionalityApi and Gpt↩ WakeupSourceRef are enabled, the driver shall report the wakeup event to EcuM through EcuM_SetWakeup↩ Event(Source) upon the hw source event.

# Chapter 5

# Module requirements

- Exclusive areas to be defined in BSW scheduler
- Exclusive areas not available on this platform
- Peripheral Hardware Requirements
- ISR to configure within AutosarOS - dependencies
- ISR Macro
- Other AUTOSAR modules - dependencies
- Data Cache Restrictions
- User Mode support
- Multicore support

## 5.1 Exclusive areas to be defined in BSW scheduler

In the current implementation, GPT is using the services of Schedule Manager (SchM) for entering and exiting the exclusive areas. The following critical regions are used in the GPT driver:

**GPT_EXCLUSIVE_AREA_00** is used in function Gpt_Init to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_00** is used in function Gpt_StartTimer to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_00** is used in function Gpt_StopTimer to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_01** is used in function Gpt_Channel_EnableChainMode to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_01** is used in function Gpt_Channel_DisableChainMode to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_02** is used in function Gpt_Init to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_02** is used in function Gpt_DisableNotification to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_02** is used in function Gpt_EnableNotification to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_02** is used in function Gpt_SetMode to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_02** is used in function Gpt_EnableWakeup to protect the TCTRL register from read/modify/write.

**GPT_EXCLUSIVE_AREA_03** is used in function Gpt_Init to protect the TFLG register from read/modify/write.

**GPT_EXCLUSIVE_AREA_03** is used in function Gpt_DisableNotification to protect the TFLG register from read/modify/write.

**GPT_EXCLUSIVE_AREA_03** is used in function Gpt_EnableNotification to protect the TFLG register from read/modify/write.

**GPT_EXCLUSIVE_AREA_04** is used in function Gpt_StartTimer to to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_05** is used in function Gpt_Init to protect the RTCS register from read/modify/write.

**GPT_EXCLUSIVE_AREA_05** is used in function Gpt_DeInit to protect the RTCS register from read/modify/write.

**GPT_EXCLUSIVE_AREA_05** is used in function Gpt_StartTimer to protect the RTCS register from read/modify/write.

**GPT_EXCLUSIVE_AREA_05** is used in function Gpt_StopTimer to protect the RTCS register from read/modify/write.

**GPT_EXCLUSIVE_AREA_06** is used in function Gpt_StartTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_06** is used in function Gpt_StopTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_07** is used in function Gpt_StopTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_07** is used in function Gpt_StartTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_10** is used in function Gpt_StartTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_10** is used in function Gpt_EnableNotification to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_10** is used in function Gpt_DisableNotification to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_10** is used in function Gpt_SetMode to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_10** is used in function Gpt_EnableWakeup to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_11** is used in function Gpt_StartTimer to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_11** is used in function Gpt_EnableNotification to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_11** is used in function Gpt_DisableNotification to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_11** is used in function Gpt_SetMode to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_12** is used in function Gpt_Init to protect the RTCC register from read/modify/write.

**GPT_EXCLUSIVE_AREA_29** is used in function Gpt_Init to protect the STM_CR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_30** is used in function Gpt_Init to protect the STM_CR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_30** is used in function Gpt_DeInit to protect the STM_CR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_30** is used in function Gpt_StartTimer to protect the STM_CR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_30** is used in function Gpt_StopTimer to protect the STM_CR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_31** is used in function Gpt_Init to protect the STM_CCR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_31** is used in function Gpt_DeInit to protect the STM_CCR register from read/modify/write.

**GPT_EXCLUSIVE_AREA_35** is used in function Gpt_Init to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_35** is used in function Gpt_EnableNotification to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_35** is used in function Gpt_DisableNotification to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_36** is used in function Gpt_Init to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_36** is used in function Gpt_StartTimer to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_36** is used in function Gpt_StopTimer to protect the UC Control register from read/modify/write.

**GPT_EXCLUSIVE_AREA_38** is used in function Gpt_StartTimer to protect the UC Control register from read/modify/write.

**Exclusive Areas implemented in Low level driver layer (IPL)**

**GPT_EXCLUSIVE_AREA_00** is used in function Pit_Ip_EnableTimer to protect the updates for:

- PIT_TCTRL

**GPT_EXCLUSIVE_AREA_01** is used in function Pit_Ip_SetChainMode to protect the updates for:

- PIT_TCTRL

**GPT_EXCLUSIVE_AREA_02** is used in function Pit_Ip_EnableInterrupt to protect the updates for:

- PIT_TCTRL

**GPT_EXCLUSIVE_AREA_03** is used in function Pit_Ip_ClearInterruptFlag to protect the updates for:

- PIT_TFLG

**GPT_EXCLUSIVE_AREA_04** is used in function Rtc_Ip_EnableCounter to protect the updates for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_05** is used in function Rtc_Ip_ClearStatusFlags to protect the updates for:

- RTC_RTCS

**GPT_EXCLUSIVE_AREA_06** is used in function Rtc_Ip_ApiEnableInterrupt to protect the updates for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_07** is used in function Rtc_Ip_ApiInterruptEnable to protect the updates for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_08** is used in function Rtc_Ip_GetTimeDate to protect the variables for:

- Rtc_Ip_u32ChState[].startTime

**GPT_EXCLUSIVE_AREA_09** is used in function Rtc_Ip_ConfigureAlarm to protect the updates for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_10** is used in function Rtc_Ip_RtcInterruptEnable to protect the variables for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_11** is used in function Rtc_Ip_RollOverInterruptEnable to protect the variables for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_12** is used in function Rtc_Ip_TriggerEnable to protect the variables for:

- RTC_RTCC

**GPT_EXCLUSIVE_AREA_35** is used in function Emios_Gpt_Ip_FlagEnable to protect the updates for:

- UC.C

**GPT_EXCLUSIVE_AREA_36** is used in function Emios_Gpt_Ip_ConfigureChannel to protect the updates for:

- UC.C

**GPT_EXCLUSIVE_AREA_38** is used in function Emios_Gpt_Ip_SetUCPEN to protect the updates for:

- UC.C

**GPT_EXCLUSIVE_AREA_29** is used in function Stm_Ip_SetDebugMode to protect the updates for:

- STM_CR

**GPT_EXCLUSIVE_AREA_30** is used in function Stm_Ip_TimerEnable to protect the updates for:

- STM_CR

**GPT_EXCLUSIVE_AREA_31** is used in function Stm_Ip_ChannelEnable to protect the updates for:

- STM_CCR

| Exclusive Area Matrix | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Exclusive Area ID | GPT_EXCLUSIVE_AREA_00 | GPT_EXCLUSIVE_AREA_01 | GPT_EXCLUSIVE_AREA_02 | GPT_EXCLUSIVE_AREA_03 | GPT_EXCLUSIVE_AREA_04 | GPT_EXCLUSIVE_AREA_05 | GPT_EXCLUSIVE_AREA_06 | GPT_EXCLUSIVE_AREA_07 | GPT_EXCLUSIVE_AREA_08 | GPT_EXCLUSIVE_AREA_09 | GPT_EXCLUSIVE_AREA_10 | GPT_EXCLUSIVE_AREA_11 | GPT_EXCLUSIVE_AREA_12 | GPT_EXCLUSIVE_AREA_38 | GPT_EXCLUSIVE_AREA_35 | GPT_EXCLUSIVE_AREA_36 | GPT_EXCLUSIVE_AREA_20 | GPT_EXCLUSIVE_AREA_21 | GPT_EXCLUSIVE_AREA_29 | GPT_EXCLUSIVE_AREA_30 | GPT_EXCLUSIVE_AREA_31 |
| GPT_EXCLUSIVE_AREA_00 | x | | | | | | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_01 | | x | | | | | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_02 | | | x | | | | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_03 | | | | x | | | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_04 | | | | | x | | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_05 | | | | | | x | | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_06 | | | | | | | x | | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_07 | | | | | | | | x | | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_08 | | | | | | | | | x | | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_09 | | | | | | | | | | x | | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_10 | | | | | | | | | | | x | | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_11 | | | | | | | | | | | | x | | | | | | | | | |
| GPT_EXCLUSIVE_AREA_12 | | | | | | | | | | | | | x | | | | | | | | |
| GPT_EXCLUSIVE_AREA_18 | | | | | | | | | | | | | | | | x | | | | | |
| GPT_EXCLUSIVE_AREA_20 | | | | | | | | | | | | | | | | | x | | | | |
| GPT_EXCLUSIVE_AREA_21 | | | | | | | | | | | | | | | | | | x | | | |
| GPT_EXCLUSIVE_AREA_29 | | | | | | | | | | | | | | | | | | | x | | |
| GPT_EXCLUSIVE_AREA_30 | | | | | | | | | | | | | | | | | | | | x | |
| GPT_EXCLUSIVE_AREA_31 | | | | | | | | | | | | | | | | | | | | | x |
| GPT_EXCLUSIVE_AREA_35 | | | | | | | | | | | | | | | x | | | | | | |
| GPT_EXCLUSIVE_AREA_36 | | | | | | | | | | | | | | | | x | | | | | |
| GPT_EXCLUSIVE_AREA_38 | | | | | | | | | | | | | | x | | | | | | | |

**Figure 5.1 Exclusive areas to be defined**

The critical regions from interrupts are grouped in “Interrupt Service Routines Critical Regions (composed diagram)”. If an exclusive area is “exclusive” with the composed “Interrupt Service Routines Critical Regions (composed diagram)” group, it means that it is exclusive with each one of the ISR critical regions.

## 5.2 Exclusive areas not available on this platform

List of exclusive areas which are not available on this platform (or blank if they're all available).

None.

None.

## 5.3 Peripheral Hardware Requirements

Driver implements channels on S32K3xx peripherals :

- S32K312:
  - PIT IP: 2 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 1 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 1 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K314:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 2 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K322:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 1 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K324:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 2 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K341:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 2 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K342:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 1 instance x 5 channels
  - RTC IP: 1 instance x 1 channel
- S32K344:
  - PIT IP: 3 instances x 4 channels (for instance 0 are 4 ch + one RTI channel)
  - STM IP: 2 instance x 4 channels
  - EMIOS IP: 1 instance x 12 channels , 2 instance x 5 channels
  - RTC IP: 1 instance x 1 channel

## 5.4 ISR to configure within AutosarOS - dependencies

isr to configure within AutosarOS dependencies template.

| ISR Name | HW INT Vector | Observations |
|---|---|---|
| ISR(PIT_0_ISR) | 96 | Interrupt for channel 0 to channel 4 |
| ISR(PIT_1_ISR) | 97 | Interrupt for channel 0 to channel 3 |
| ISR(PIT_2_ISR) | 98 | Interrupt for channel 0 to channel 3 |
| ISR(RTC_0_Ch_0_ISR) | 102 | RTCF or ROVRF interrupt to be serviced by the system controller |
| ISR(STM_0_ISR) | 24 | Single interrupt vector for all four channels |
| ISR(STM_1_ISR) | 25 | Single interrupt vector for all four channels |
| (EMIOS_0_CH_0) | 66 | Interrupt request for channel |
| (EMIOS_0_CH_1) | 66 | Interrupt request for channel |
| (EMIOS_0_CH_2) | 66 | Interrupt request for channel |
| (EMIOS_0_CH_3) | 66 | Interrupt request for channel |
| (EMIOS_0_CH_4) | 65 | Interrupt request for channel |
| (EMIOS_0_CH_5) | 65 | Interrupt request for channel |
| (EMIOS_0_CH_6) | 65 | Interrupt request for channel |
| (EMIOS_0_CH_7) | 65 | Interrupt request for channel |
| (EMIOS_0_CH_8) | 64 | Interrupt request for channel |
| (EMIOS_0_CH_9) | 64 | Interrupt request for channel |
| (EMIOS_0_CH_10) | 64 | Interrupt request for channel |
| (EMIOS_0_CH_11) | 64 | Interrupt request for channel |
| (EMIOS_0_CH_12) | 63 | Interrupt request for channel |
| (EMIOS_0_CH_13) | 63 | Interrupt request for channel |
| (EMIOS_0_CH_14) | 63 | Interrupt request for channel |
| (EMIOS_0_CH_15) | 63 | Interrupt request for channel |
| (EMIOS_0_CH_16) | 62 | Interrupt request for channel |
| (EMIOS_0_CH_17) | 62 | Interrupt request for channel |
| (EMIOS_0_CH_18) | 62 | Interrupt request for channel |
| (EMIOS_0_CH_19) | 62 | Interrupt request for channel |
| (EMIOS_0_CH_20) | 61 | Interrupt request for channel |
| (EMIOS_0_CH_21) | 61 | Interrupt request for channel |
| (EMIOS_0_CH_22) | 61 | Interrupt request for channel |
| (EMIOS_0_CH_23) | 61 | Interrupt request for channel |
| (EMIOS_1_CH_0) | 74 | Interrupt request for channel |
| (EMIOS_1_CH_1) | 74 | Interrupt request for channel |
| (EMIOS_1_CH_2) | 74 | Interrupt request for channel |
| (EMIOS_1_CH_3) | 74 | Interrupt request for channel |
| (EMIOS_1_CH_4) | 73 | Interrupt request for channel |
| (EMIOS_1_CH_5) | 73 | Interrupt request for channel |
| (EMIOS_1_CH_6) | 73 | Interrupt request for channel |
| (EMIOS_1_CH_7) | 73 | Interrupt request for channel |
| (EMIOS_1_CH_8) | 72 | Interrupt request for channel |

| ISR Name | HW INT Vector | Observations |
|---|---|---|
| (EMIOS_1_CH_9) | 72 | Interrupt request for channel |
| (EMIOS_1_CH_10) | 72 | Interrupt request for channel |
| (EMIOS_1_CH_11) | 72 | Interrupt request for channel |
| (EMIOS_1_CH_12) | 71 | Interrupt request for channel |
| (EMIOS_1_CH_13) | 71 | Interrupt request for channel |
| (EMIOS_1_CH_14) | 71 | Interrupt request for channel |
| (EMIOS_1_CH_15) | 71 | Interrupt request for channel |
| (EMIOS_1_CH_16) | 70 | Interrupt request for channel |
| (EMIOS_1_CH_17) | 70 | Interrupt request for channel |
| (EMIOS_1_CH_18) | 70 | Interrupt request for channel |
| (EMIOS_1_CH_19) | 70 | Interrupt request for channel |
| (EMIOS_1_CH_20) | 69 | Interrupt request for channel |
| (EMIOS_1_CH_21) | 69 | Interrupt request for channel |
| (EMIOS_1_CH_22) | 69 | Interrupt request for channel |
| (EMIOS_1_CH_23) | 69 | Interrupt request for channel |
| (EMIOS_2_CH_0) | 82 | Interrupt request for channel |
| (EMIOS_2_CH_1) | 82 | Interrupt request for channel |
| (EMIOS_2_CH_2) | 82 | Interrupt request for channel |
| (EMIOS_2_CH_3) | 82 | Interrupt request for channel |
| (EMIOS_2_CH_4) | 81 | Interrupt request for channel |
| (EMIOS_2_CH_5) | 81 | Interrupt request for channel |
| (EMIOS_2_CH_6) | 81 | Interrupt request for channel |
| (EMIOS_2_CH_7) | 81 | Interrupt request for channel |
| (EMIOS_2_CH_8) | 80 | Interrupt request for channel |
| (EMIOS_2_CH_9) | 80 | Interrupt request for channel |
| (EMIOS_2_CH_10) | 80 | Interrupt request for channel |
| (EMIOS_2_CH_11) | 80 | Interrupt request for channel |
| (EMIOS_2_CH_12) | 79 | Interrupt request for channel |
| (EMIOS_2_CH_13) | 79 | Interrupt request for channel |
| (EMIOS_2_CH_14) | 79 | Interrupt request for channel |
| (EMIOS_2_CH_15) | 79 | Interrupt request for channel |
| (EMIOS_2_CH_16) | 78 | Interrupt request for channel |
| (EMIOS_2_CH_17) | 78 | Interrupt request for channel |
| (EMIOS_2_CH_18) | 78 | Interrupt request for channel |
| (EMIOS_2_CH_19) | 78 | Interrupt request for channel |
| (EMIOS_2_CH_20) | 77 | Interrupt request for channel |
| (EMIOS_2_CH_21) | 77 | Interrupt request for channel |
| (EMIOS_2_CH_22) | 77 | Interrupt request for channel |
| (EMIOS_2_CH_23) | 77 | Interrupt request for channel |

## 5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

### 5.5.1 Without an Operating System

The macro *USING_OS_AUTOSAROS* must not be defined.

#### 5.5.1.1 Using Software Vector Mode

The macro *USE_SW_VECTOR_MODE* must be defined and the ISR macro is defined as:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

#### 5.5.1.2 Using Hardware Vector Mode

The macro *USE_SW_VECTOR_MODE* must not defined and the ISR macro is defined as:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, the drivers' interrupt handlers must also handle the context save and restore.

### 5.5.2 With an Operating System

Please refer to your OS documentation for description of the ISR macro.

## 5.6 Other AUTOSAR modules - dependencies

- BASE: The BASE module contains the common files/definitions needed by all MCAL modules.
- DET - Development Error Tracer: This module is necessary for enabling Development error detection. The API function used is Det_ReportError(). Activation of Development error detection is configurable using 'GptDevErrorDetect' configuration parameter.
- DEM - Diagnostic Event Manager: This module is necessary for enabling reporting of production relevant error status. It is not used with current GPT implementation as the production relevant error codes are not present.
- EcuC module: This module is necessary for handling PostBuild Variant. It allows users to configure multiple configuration.
- RESOURCE: The RESOURCE module is used to select microcontroller's derivatives.

## 5.7 Data Cache Restrictions

None.

## 5.8 User Mode support

- User Mode configuration in the module
- User Mode configuration in AutosarOS

### 5.8.1 User Mode configuration in the module

The GPT can be run in user mode if the following steps are performed:

- Enable **GptEnableUserModeSupport** from the configuration
- Call the following functions as trusted functions:

| Function syntax | Available via | Description |
|---|---|---|
| void Stm_Ip_SetUserAccess↩ Allowed(uint32 StmBaseAddr) | Stm_Ip_Trusted↩ Functions.h | For seting the user access allowed for System Timer Module registers protected by REG_PROT |

### 5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may has the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header `<IpName>_Ip↩ _TrustedFunctions.h`. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter User Mode configuration in the module for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

`#define` OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)

So, the following steps need to be done in AutosarOS:

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will unmarshalling of the parameters to call `<Function_Name>`() of driver. The `<Function_Name>`() functions are already defined in driver and declared in `<IpName>_Ip_TrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

See the sequence chart below for an example calling `Linflexd_Uart_Ip_Init_Privileged()` as a trusted function.

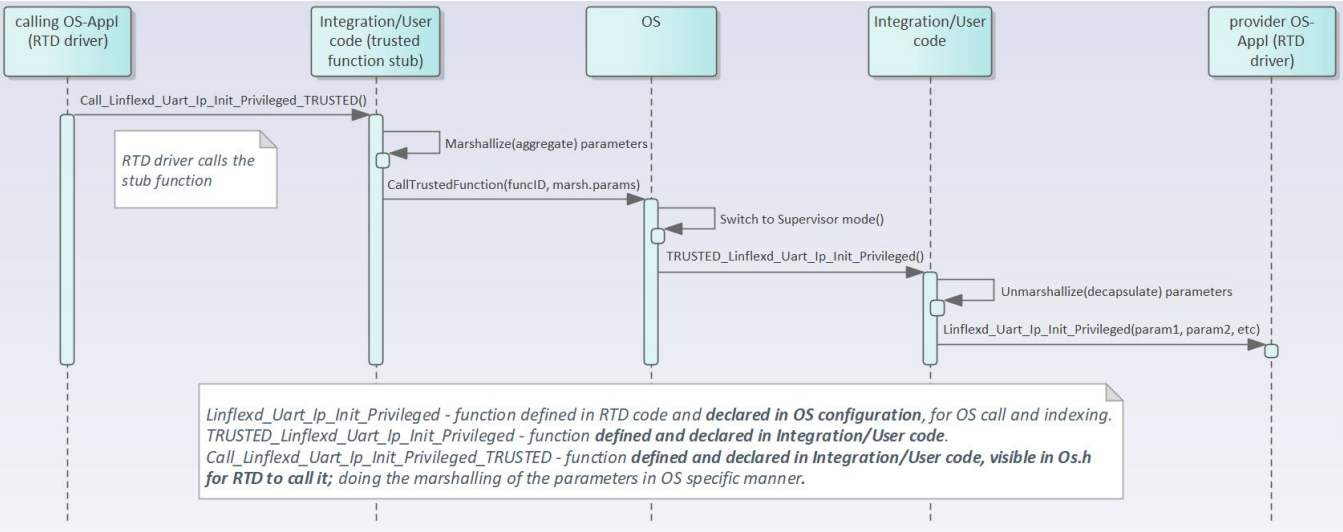



**Figure 5.2 Example sequence chart for calling `Linflexd_Uart_Ip_Init_Privileged` as trusted function**

## 5.9 Multicore support

1.** The Gpt implements the "Autosar 4.4 MCAL Multicore Distribution" according to type II, in which the mappable element is set to Hw_Unit for PIT Ip, STM Ip , RTC Ip and EMIOS Ip. For additional details, please refer to AUTOSAR_EXP_BSWDistributionGuide.

2.** The Gpt and the mappable elements can be allocated to zero, one or several ECUC partitions, by means of "GptEcucPartionRef". If the Gpt is mapped to zero ECUC partitions, the Gpt behavior reverts to single-core implementation, similar to previous Autosar versions. If the Gpt is mapped to one or more ECUC partitions, the Gpt enforces the following multi-core assumptions:

The Gpt assumes there is a single EcucPartition allocated per core. Internally, the module will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements. The Gpt assumes the EcucCoreIDs are defined in a compact/consecutive order, starting from zero. The rationale is that the number of EcucPartitions is used for dimensioning the Gpt internal variables and the EcucCoreIDs are used for indexing those variables. The Gpt assumes that initialization is performed on each core, Gpt_Init() is called separately for each core, using a different configguration structure. (Type II) The Gpt initialization expects the upper layer will pass the correct initialization pointer, specific to the partition in which the driver is to be used. For example: EcucPartition_1 is assigned to CoreID 1; Gpt_Init function will be called with Gpt_Config_EcucPartition_1 configuration structure, on Core 1. The Gpt will check upon each API call if the requested resource is configured to be available on the current core, if DET error reporting is enabled. The Gpt requires that all variables in NonCacheable MemMap sections be allocated accordingly, to avoid data corruption in multicore context. The Gpt assumes that RTE module implements the EXCLUSIVE AREAS to be core-aware only. The rationale is that the module implementation ensures data integrity by separating the mappable elements for different cores already, thus implementing the EX↩ CLUSIVE AREAS in a blocking manner (ex: spin-lock) on a multicore scope, might affect the performance of the drivers on the two cores, although they might access separate HW elements. For single-core scope, the EXCLUSIVE AREAS keep the same purpose as on previous AUTOSAR implementations. The Gpt assumes that each interrupt

is routed by the system only to the core on which is supposed to be serviced. The configuration structure name shall be available in the caller scope of Gpt_Init function by being declared with EXTERN, according to its generated name.

Module specific limitation:**

For current implementation, Gpt driver does not support channel mapping with zero ECUC partition in configuration. Therefore the driver will force user to map each channel with one partition only.

Current implementation of GPT driver does not allow for a partition with no allocated channels to access the predefined timer info – if any of the predefined timers are defined. The GPT driver issues a DET for this access if the API call is present in user code

# Chapter 6

# Main API Requirements


- Main function calls within BSW scheduler
- API Requirements
- Calls to Notification Functions, Callbacks, Callouts


## 6.1 Main function calls within BSW scheduler

None.

## 6.2 API Requirements

None.

## 6.3 Calls to Notification Functions, Callbacks, Callouts

**Call-back Notifications:**

- There are no call-back notifications defined inside the GPT driver.

**User Notification:**

- The GPT Driver provides a notification per channel that is called whenever the defined time period is over.
- The notifications can be configured as pointers to user defined functions. If notification is not desired,

'NULL_PTR' shall be configured.

An example of the syntax of this function is as follows:

- void Gpt_Notification_<channel>(void)
- An extern declaration of this function is available in Gpt_PBcfg.c. The function has to be implemented by the user.

# Chapter 7

# Memory allocation


- Sections to be defined in Gpt_MemMap.h
- Linker command file


## 7.1 Sections to be defined in Gpt_MemMap.h

| Section name | Type of section | Description |
|---|---|---|
| GPT_START_SEC_CONFIG_DATA_UNSPECIFIED | Configuration Data | Start of Memory Section for Config Data |
| GPT_STOP_SEC_CONFIG_DATA_UNSPECIFIED | Configuration Data | End of memory Section for Config Data |
| GPT_START_SEC_CODE | Code | Start of memory Section for Code |
| GPT_STOP_SEC_CODE | Code | End of memory Section for Code |
| GPT_START_SEC_VAR_INIT_UNSPECIFIED | Variables | Start of memory Section for Variables |
| GPT_STOP_SEC_VAR_INIT_UNSPECIFIED | Variables | End of memory Section for Variables |
| GPT_START_SEC_VAR_CLEARED_UNSPECIFIED | Variables | Start of memory Section for Variables. Used for variables, constants, structure, array and unions when SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. For instance used for variables of unknown size |
| GPT_STOP_SEC_VAR_CLEARED_UNSPECIFIED | Variables | End of memory Section for Variables. Used for variables, constants, structure, array and unions when SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. For instance used for variables of unknown size |

## 7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>"_MemMap.h.

# Chapter 8

# Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section Files Required for Compilation
2. Allocate the proper memory sections in the driver's memory map header file ("<Module>"_MemMap.h) and linker command file. For more details refer to section Sections to be defined in <Module>_MemMap.h
3. Compile & build the module with all the dependent modules. For more details refer to section Building the Driver

# Chapter 9

# External assumptions for driver

The section presents requirements that must be complied with when integrating the GPT driver into the application.

| External Assumption Req ID | External Assumption Text |
|---|---|
| SWS_Gpt_00353 | If the register can affect several hardware modules and if it is an I/O register it shall be initialized by the PORT driver. Note: The GPT driver manages hardware which does not include input/output configurable pins. |
| SWS_Gpt_00354 | If the register can affect several hardware modules and if it is not an I/O register it shall be initialized by the MCU driver. Note: The requirement is implicitly fulfilled at MCU level, as the MCU shall initialize the clock tree used also by the GPT driver. |
| SWS_Gpt_00355 | One-time writable registers that require initialization directly after reset shall be initialized by the startup code. Note: to be traced at the sMCAL generic level; no verification done on driver level.The Interrupt Controller shall be initialized by the integrating application before to start using the GPTdriver. |
| SWS_Gpt_00356 | All other registers shall be initialized by the startup code. Note: to be traced at the sMCAL generic level; no verification done on driver level.The Interrupt Controller shall be initialized by the integrating application before to start using the GPTdriver. |
| EA_RTD_00032 | The application shall not preempt a channel related function (like starting/stoping a timer) by calling Gpt_SetMode() or Gpt_DeInit(). |
| EA_RTD_00033 | The application shall not preempt a GPT function working on a GPT channel by calling another GPT function targeting the same channel. |
| EA_RTD_00034 | The application must not concurrently call Gpt functions with one exception: GetVersionInfo only can get interrupted or may interrupt. Note: A transversal GPT functions are those functions addressing the entire set of channels, like Gpt_Init(), Gpt_DeInit(), Gpt_SetMode(), Gpt_Periodic↩ Check(), ... |
| EA_RTD_00035 | The application shall not call any function of the GPT module before having called Gpt_Init. |
| EA_RTD_00036 | Wakeup enabled timers shall be started or stopped only when GPT driver is in GPT_MODE_NORMAL mode.The external application shall invoke Gpt_EnableWakeup() and Gpt_DisableWakeup() only when GPT driver is in GPT_MODE_NORMAL mode. Note: If Gpt_EnableWakeup(), Gpt↩ _DisableWakeup(), Gpt_StartTimer() and Gpt_StopTimer() are called while GPT is already in SLEEP mode, the GPT driver behavior is not guaranteed. Therefore any wakeup channel configuration shall be done before entering in sleep mode. |

| External Assumption Req ID | External Assumption Text |
|---|---|
| EA_RTD_00071 | If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used. |
| EA_RTD_00081 | The integrator shall assure that <MSN>_Init() and <MSN>_DeInit() functions do not interrupt each other. |
| EA_RTD_00082 | When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: **Rationale**: This ensures that no other buffers/variables compete for the same cache lines. |
| EA_RTD_00092 | The integrator shall allocate a single EcucPartition per core or the partition in which the Gpt is allocated shall be exclusively mapped to a core. Note↩ : Internally, the Gpt will use the Core ID returned by GetCoreID API to reference the appropriate global data and configuration elements, that is why a core should reference only one configured partition. |
| EA_RTD_00093 | The application shall define EcucCoreIDs in a compact/consecutive order, starting from zero. |
| EA_RTD_00094 | When multicore support is enabled, the application shall call Gpt_Init() for each core, using the dedicated configuration pointer for that core. |
| EA_RTD_00096 | The application shall pass the correct initialization pointer, specific to the partition in which the driver is to be used. |
| EA_RTD_00106 | Standalone IP configuration and HL configuration of the same driver shall be done in the same project |
| EA_RTD_00107 | The integrator shall use the IP interface only for hardware resources that were configured for standalone IP usage. Note:The integrator shall not directly use the IP interface for hardware resources that were allocated to be used in HL context. |
| EA_RTD_00108 | The integrator shall use the IP interface to a build a CDD, therefore the BSWMD will not contain reference to the IP interface |

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